



CS & IT ENGINEERING



Algorithms

Dynamic Programming

Lecture No.- 05



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

0/1 Knap

LCS

Topics to be Covered



Topic

Topic

Topic

LCS

MCM



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Dynamic Programming



Longest Common Sub-Sequence (LCS):

- Based on Strings
- A sequence/~~group~~/array of one or more characters.



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String:

1) Substring:

- A contiguous sequence of 1 or more characters from the given string.



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String:

2) Subsequence:

- A sequence of 1 or more characters from the given string, that may or may not be contiguous, but their relative order is maintained.



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Example:

- String: "ABC" $\rightarrow$ length = 3
- Sub-strings: A, B, C $\rightarrow$ 3

AB, BC $\rightarrow$ 2

ABC $\rightarrow$ 1

No. of substrings = $3 + 2 + 1 = 6$



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Important Result:

Given : $S[1, \dots, n]$

#Q. What are the total number of sub-strings of at-least length 1 (non-empty sub-strings)?

$$\frac{n(n+1)}{2}$$



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Sub-Sequence:

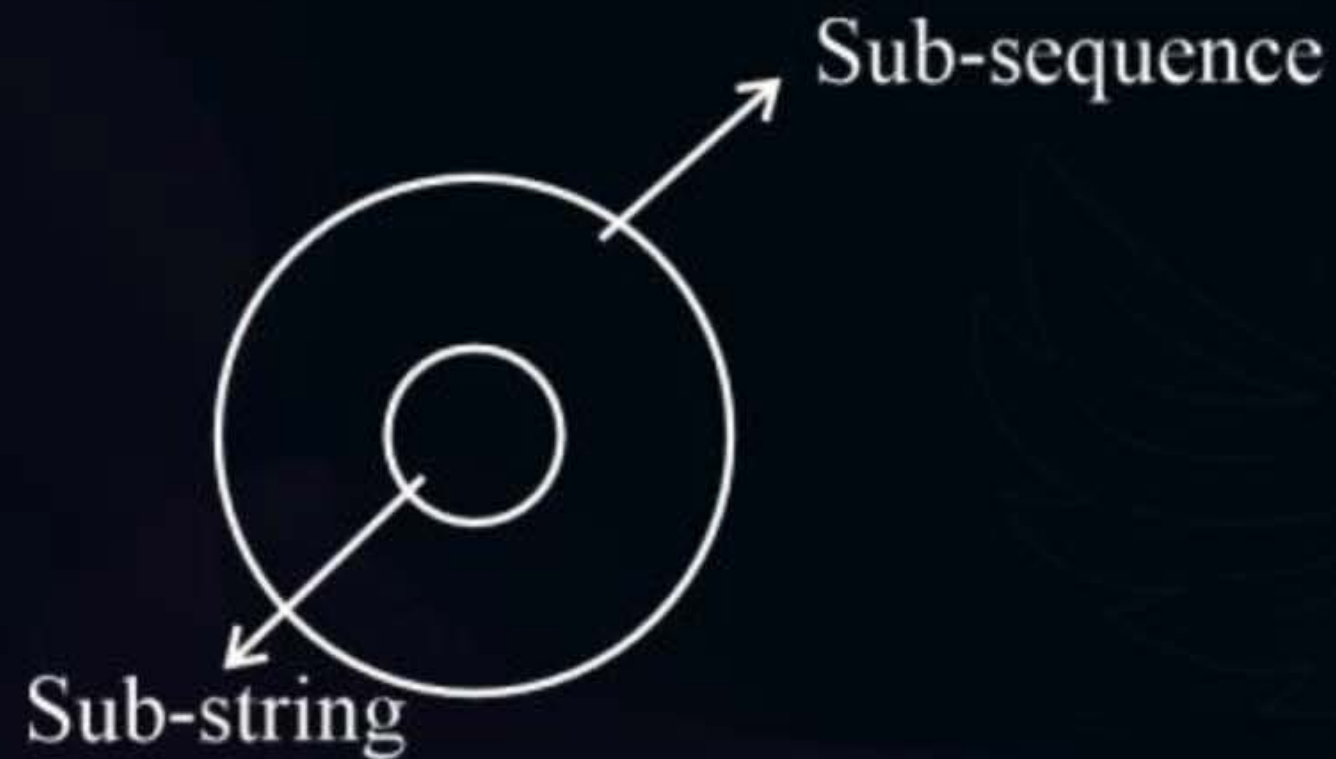
- String = "ABC"
- Sub-Sequence = A, B, C, AB, BC, AC, ABC → Valid sub-sequence
- CA, CB → relative order not maintained, not a sub-sequence.



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Important:

- Every substring is also a sub-sequence, but a Sub-sequence may or may not be a substring.



[MSQ]

#Q. Given a string $S = \text{"CABDABBBCD"}$.
Which of the following are valid sub-sequences?

A

BBD

B

BAC

C

CDCA

D

CADC

D

ADABD



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Longest Common Sub-Sequence:

#Q. Given a string of length 'n' characters, the number of sub-sequences possible are?

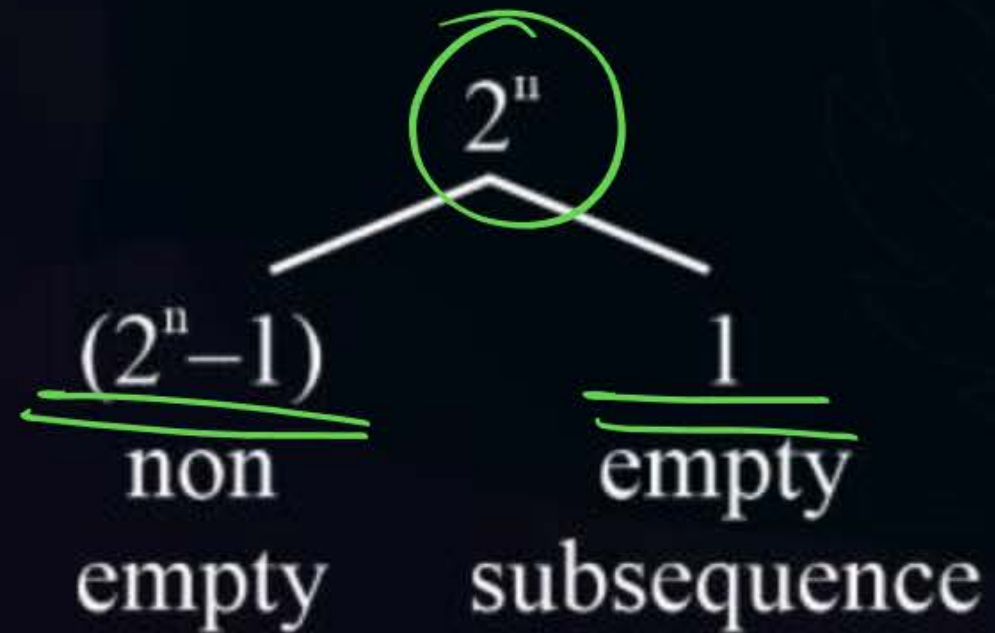


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Longest Common Sub-Sequence:



$\phi \rightarrow$ empty sub-sequence





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Brute Force/Enumeration:

All Subsequence $\rightarrow A$

All Subsequence $\rightarrow B$

- $O(2^n)$
- Hence, we need DP based approach.



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Longest Common Subsequence (DP based approach):

Common Subsequence:

- Given two strings X and Y of length n and m characters, a subsequence that is common to both X and Y is known as a common sub-sequence.



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Example:

#Q. $X = \text{"ABCD"}, Y = \text{"BDC"}$

Which sub-sequences are common to both?

1) A

2) B ✓

3) BC ✓

4) BD ✓

5) BCD

6) DC

7) BDC



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LCS Problem Statement:

#Q. Given strings 'A' and 'B' of length (Characters) 'n' & 'M' respectively. The LCS problem is to determine the subsequence that is of the longest length among all the subsequences that are common to both 'A' & 'B'?



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Example:

$S1 = \text{"ABCB DAB"}$

$S2 = \text{"BDCABA"}$

Longest LCS = 4



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Common Subsequences:

$AB \rightarrow 2$

$BCA \rightarrow 3$

$BCB \rightarrow 3$

$BCBA \rightarrow 4$

$BCAB \rightarrow 4$

- Multiple longest common subsequences are possible.
- But LCS length is unique $\rightarrow 4$



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DP based solution to LCS:

- Let 'i' and 'j' be indices in the strings 'X' and 'Y' respectively, such that.

$$X = [x_1, x_2, x_3, \dots, x_n]$$
$$Y = [y_1, y_2, y_3, \dots, y_m]$$



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- Let $L(i, j)$ represent the length of the common sub-sequence of the strings X and Y .

$$\begin{aligned} L(i, j) &= 1 + L(i-1, j-1); \text{ if } X[i] == Y[j] \\ &= \max\{L(i-1, j), L(i, j-1)\}; \text{ if } X[i] \neq Y[j] \end{aligned}$$



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Example:

Case1: $X[i] = Y[j]$

X = "ABBCDCCABA"

Y = "CBDADCBDAA"





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Example:

Case2: $X[i] \neq Y[j]$

X = "ABCDDBCCADB"

Y = "BDBC AAABDCD"

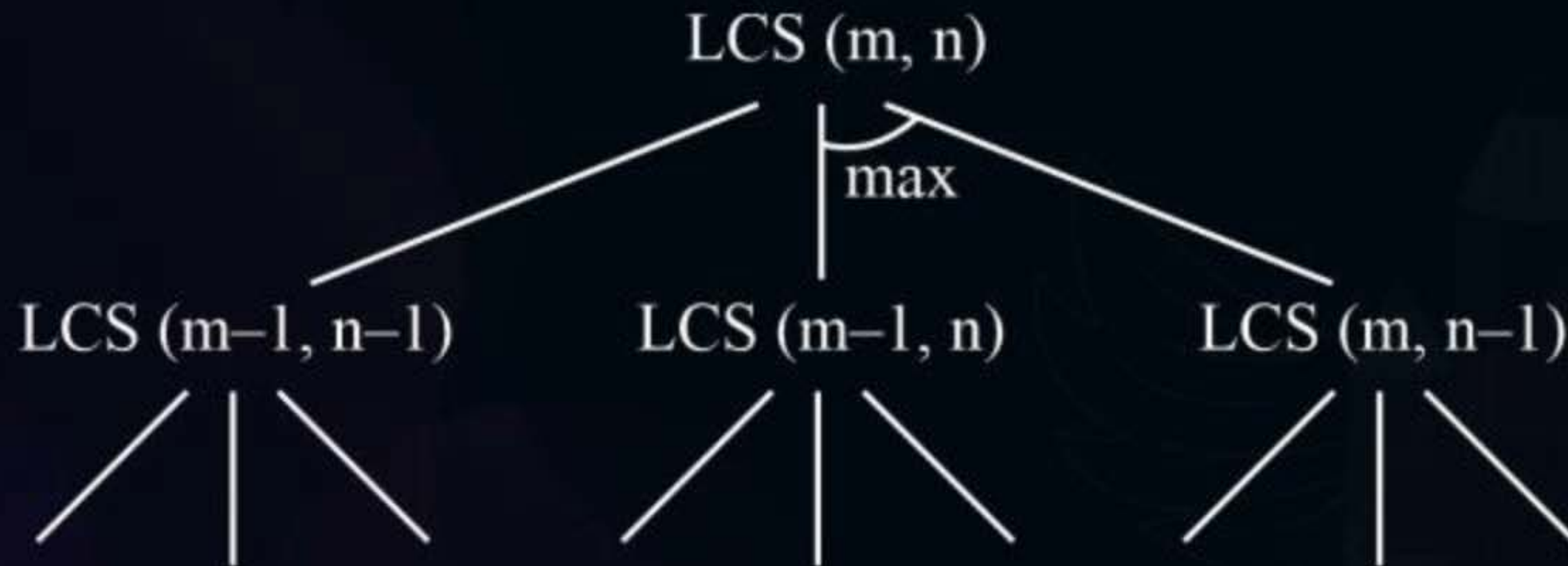
$\max\{L(i, j-1), L(i-1, j)\}$



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Overlapping Subproblems:

$$|X| = n, |Y| = m$$

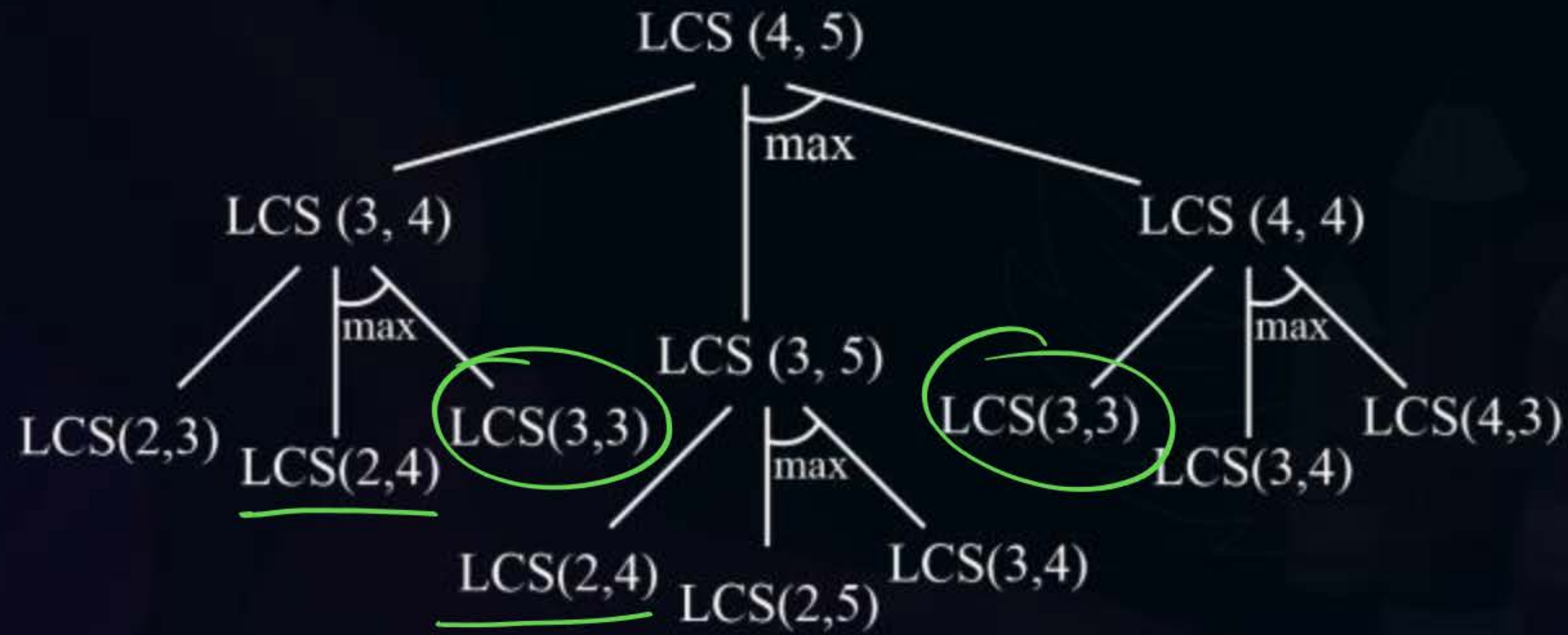




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Overlapping Subproblems:

$$|X| = 4, |Y| = 5$$





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Example:

Top-Down (Memoization/Recurrence):

X = "ABBAB"

Y = "ACBAB"

LCS : "ABAB"

Length = 4



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Example:

Bottom-up Approach (Tabulation):

X = "ABCBDAB" → length = 7

Y = "BDCABA" → length = 6

BCAB

BDAB



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X =

		0	1	2	3	4	5	6
			B	D	C	A	B	A
0		0	0	0	0	0	0	0
1	A	0	0	0	0	1	1	1
2	B	0	1	1	1	1	2	2
3	C	0	1	1	2	2	2	2
4	B	0	1	1	2	2	3	3
5	D	0	1	2	2	2	3	3
6	A	0	1	2	2	3	3	4
7	B	0	1	2	2	3	4	4



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LCS Recurrence:

- $LCS(i, j) = 1 + LCS(i-1, j-1); \text{ if } X[i] = Y[j]$
- $LCS(i, j) = \max\{L(i, j-1), L(i-1, j)\}; \text{ if } X[i] \neq Y[j]$



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Algorithm LCS based on Bottom-up Tabulation method :

Algorithm LCS (x, y)

Integer x[0..n], y [0..m];

{

integer L [0..n , 0..m];

1. for i $\leftarrow$ 0 to n - 1

L [i, - 1] = 0;

2. for j $\leftarrow$ 0 to m - 1

L [-1 , j] = 0;

3. for i $\leftarrow$ 0 to n - 1

for j $\leftarrow$ 0 to m - 1



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if ($x[i] = y[j]$) then

$$L[i, j] = 1 + L[i - 1, j - 1];$$

else

$$L[i, j] = \max \{L[i, j - 1], L[i - 1, j]\};$$

gym



gym

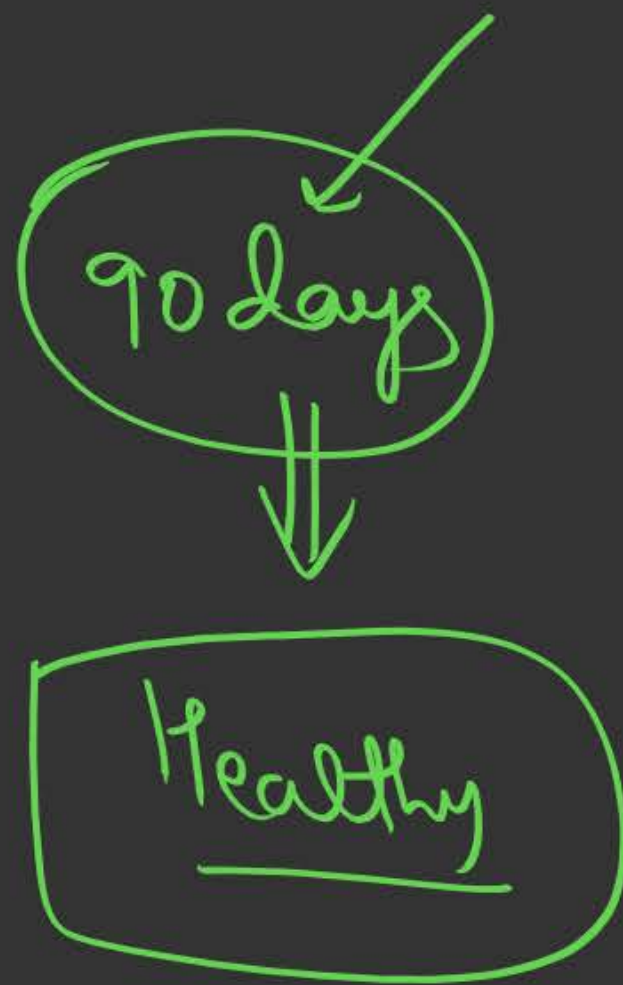
Focus → process

15 hrs/day → 2 days



30 hrs

1 hr/day → 30 days → 30 hrs



Plan

✓ $m_1, m_2, \dots, m_n$

July $\xrightarrow{\text{Aug Sep}}$ Jan

Monthly

Buffer

5 days

W1

w_2

W3

W4

11

D2

D4

D5

06

mcm



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Matrix Chain Multiplication/Product (MCM):

#Q. Given two matrices A and B of size $m \times n$ and $n \times p$.

How many scalar multiplication are required in $A_{m \times n} * B_{n \times p} = C$?



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Case 1:

Square Matrices:

$$A_{n \times n} * B_{n \times n} = C_{n \times n}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} * \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$



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$$c_{11} = a_{11} \times b_{11} + a_{12} \times b_{21}$$

$$c_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$$

$$c_{21} = a_{21} \times b_{11} + a_{22} \times b_{21}$$

$$c_{22} = a_{21} \times b_{12} + a_{22} \times b_{22}$$

Every element = 2

Hence, total scalar multiplication = $2 \times (2 \times 2) = 2 \times 2 \times 2 = 8$



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Case 2:

Non-Square Matrices:

$$A_{m \times n} * B_{n \times p} = \underline{\underline{C_{m \times p}}}$$

- Hence, for all $m \times p$ element.

$$\text{Total scalar multiplication} = n \times (m \times p) = \boxed{m \times n \times p}$$



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Important Result:

$$A_{m \times n} * B_{n \times p} = C_{m \times p}$$

- Total number of scalar multiplication = $m \times n \times p$



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MCM Problem Statement:

- Given a chain of ^{compatible} ~~computable~~ non-square matrices, it is required to multiply them together to get a final resultant matrix.

with min scalar multi



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Example:

$$\underline{\underline{n = 3}}$$

$$[A_{2 \times 10}, B_{10 \times 50}, C_{50 \times 20}] = \underline{\underline{(ABC)}} = Z_{2 \times 20}$$

min



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Parenthesizing Problem:

$$ABC = (1) \underline{(A * B)} * C \Rightarrow 3000$$

$$(2) A * \underline{(B * C)} \Rightarrow 10,400$$



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(1) $(A * B) * C$

How many Scalar multiplication?

(i) $(A * B) = 2 \times 10 \times 50$

(ii) $\overset{R}{\underbrace{(A * B)} * \underbrace{C}} = \underbrace{2 \times 50 \times 20}$

So, total scalar multiplication for $(A * B) * C$ are

$$= \underbrace{2 \times 10 \times 50} + \underbrace{2 \times 50 \times 20}$$

$$= 1000 + 2000 = \textcircled{3000}$$



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(2) $A * (B * C)$

(i) $(B * C) = \underline{10 \times 50 \times 20}$

(ii) $A * (B * C) = \underline{2 \times 10 \times 20}$

So, total scalar multiplication for $A * (B * C)$ are

$$= \underline{10 \times 50 \times 20} + \underline{2 \times 10 \times 20}$$

$$= 1000 + 400$$

$$= \underline{10400}$$

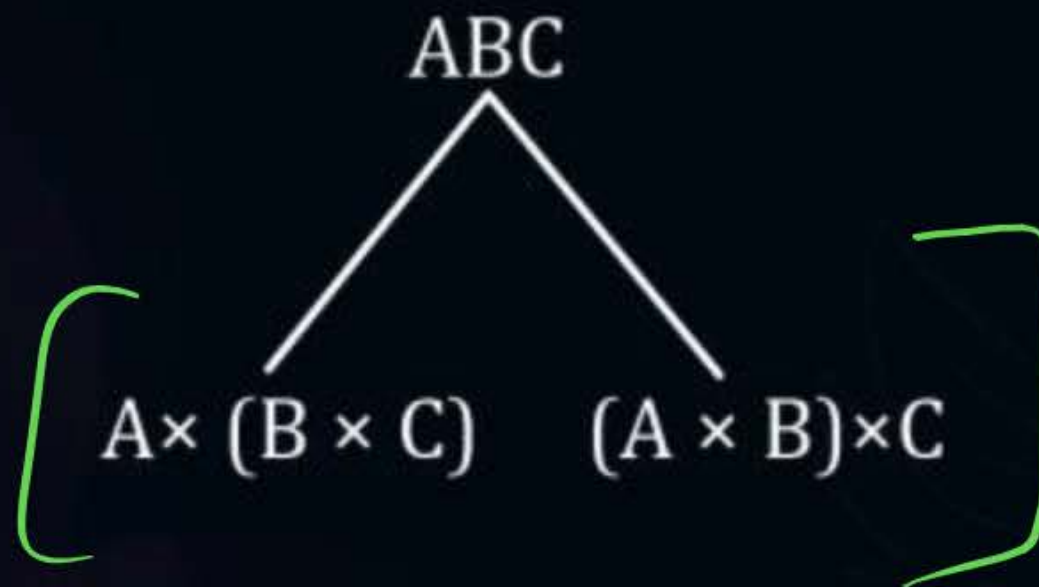


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MCM objective function

to minimize the number of scalar multiplication.





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DP based approach:-

- Given a chain of n matrixes $(A_1, A_2, \dots, A_n)$ where matrix A_i is of Dimension

$$P_{i-1} * P_i.$$

The MCM problem is to fully parenthesize the chain such that the total number of scalar multiplication are minimized.





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#Q. How many ways to multiply chain n matrices?

K = 1

K = 2

ABC

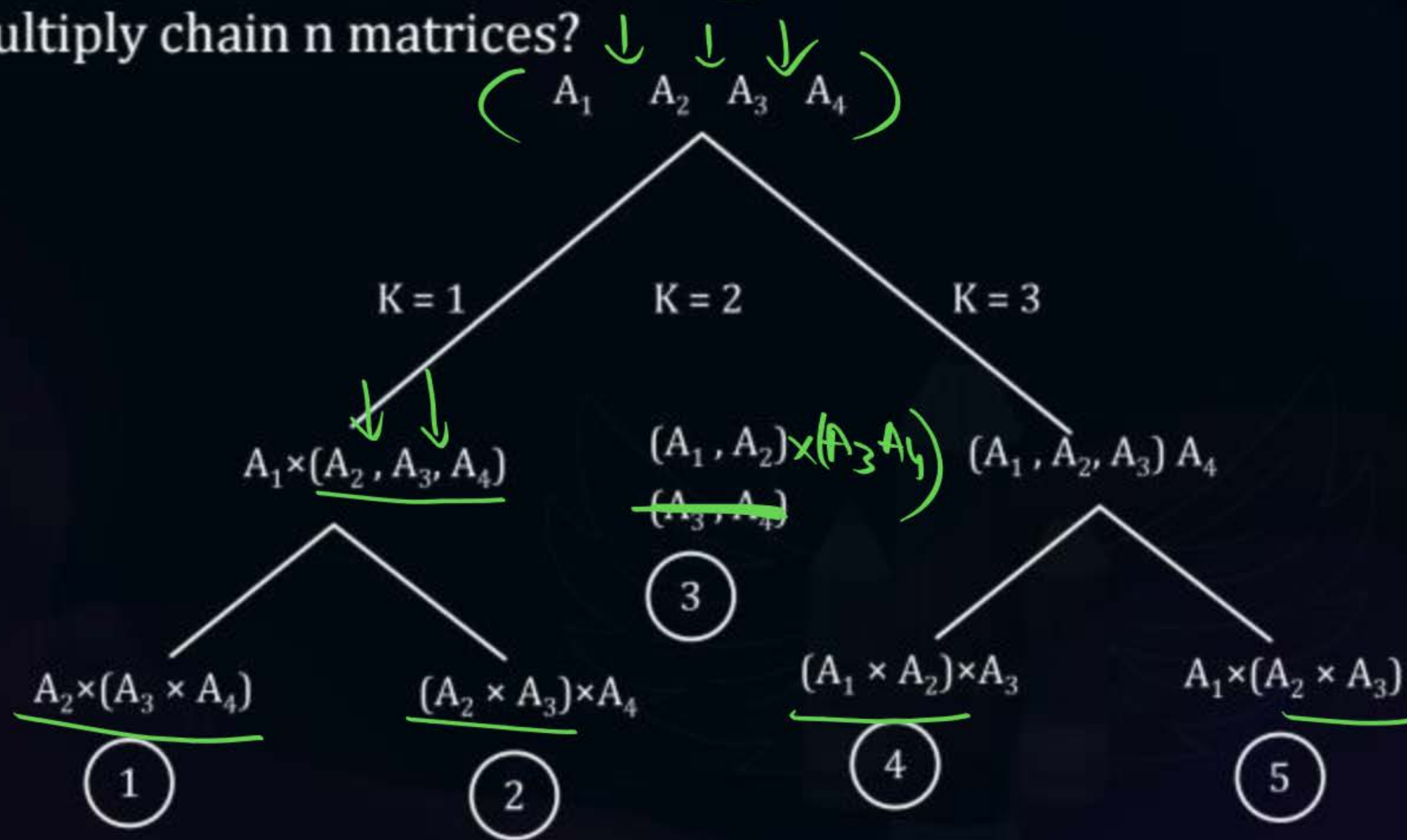
(1) $A*(B*C)$

(2) $(B*C)*C$

$(A*B)*C$

3

V.V.IMP





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Generalized Results:-

Total no. of ways to multiply a chain of $(n + 1)$ matrix $\Rightarrow$ Catalan number C_n

E.g. $N = 3$

↓

4 matrix

$$= \frac{1}{4} \times {}^6C_3$$

$$= \frac{1}{4} \times \frac{6 \times 5 \times 4 \times 3}{3! \times 3!}$$

$$= \frac{6 \times 5}{5} = 5$$

$n+1$ mat $\rightarrow C_n$

$$\boxed{\frac{1}{(n+1)} \times {}^{2n}C_n}$$

$n+1$
 $n \Rightarrow n-1$



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$$\underline{\underline{n \text{ Matrix}}} \rightarrow \frac{1}{n} \times 2^{(n-1)} C_{(n-1)}$$

$$\underline{\underline{Cat(n-1)}}$$



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Derivation

~~Derivation~~ of DP based Recurrence for ~~men~~ ^{mcm}:

Let the resultant matrix $A_{i j}$ be of product

$$(A_i * A_{i+1} * A_{i+2} \dots A_j)$$

Chain $\rightarrow$ Result $A_{i j}$

$$A_1 \times A_2 \times A_3 \dots A_n \Rightarrow A_{1 \times n}$$

$$\underline{\underline{k = (1 \text{ to } n-1)}}$$

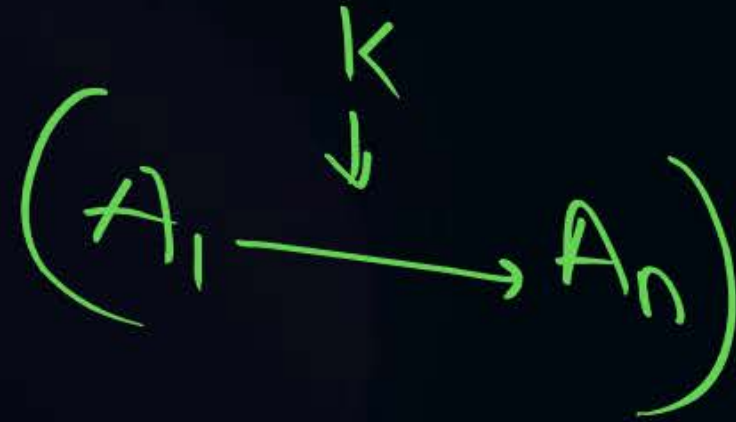
$\downarrow \quad \downarrow$
A B C



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Any optimal Parameterization must split the chain about the matrix A_k & A_{k+1} ^{Such} ~~rule~~ → that the number of scalar multiplications are minimum.



$$\underline{1 \leq k \leq (n-1)}$$



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DP Recurrence

Let $m[i, j]$ represent the number of scalar multiplication to get the resultant matrix $A_i j$.

$$[A_i \times A_{i+1} \dots\dots\dots A_k * \dots\dots\dots A_j] \Rightarrow \underline{A_i j} \rightarrow \underline{M[i, j]}$$

Split at k

$$(\underbrace{A_i \times A_{i+1} + \dots\dots\dots A_k}_{m[i, k]}) \times (\underbrace{A_{k+1} \dots\dots\dots A_j}_{m[k+1, j]}) \quad \text{Where } \rightarrow \underline{i \leq k \leq (j-1)}$$

$m[i, k]$

$m[k+1, j]$

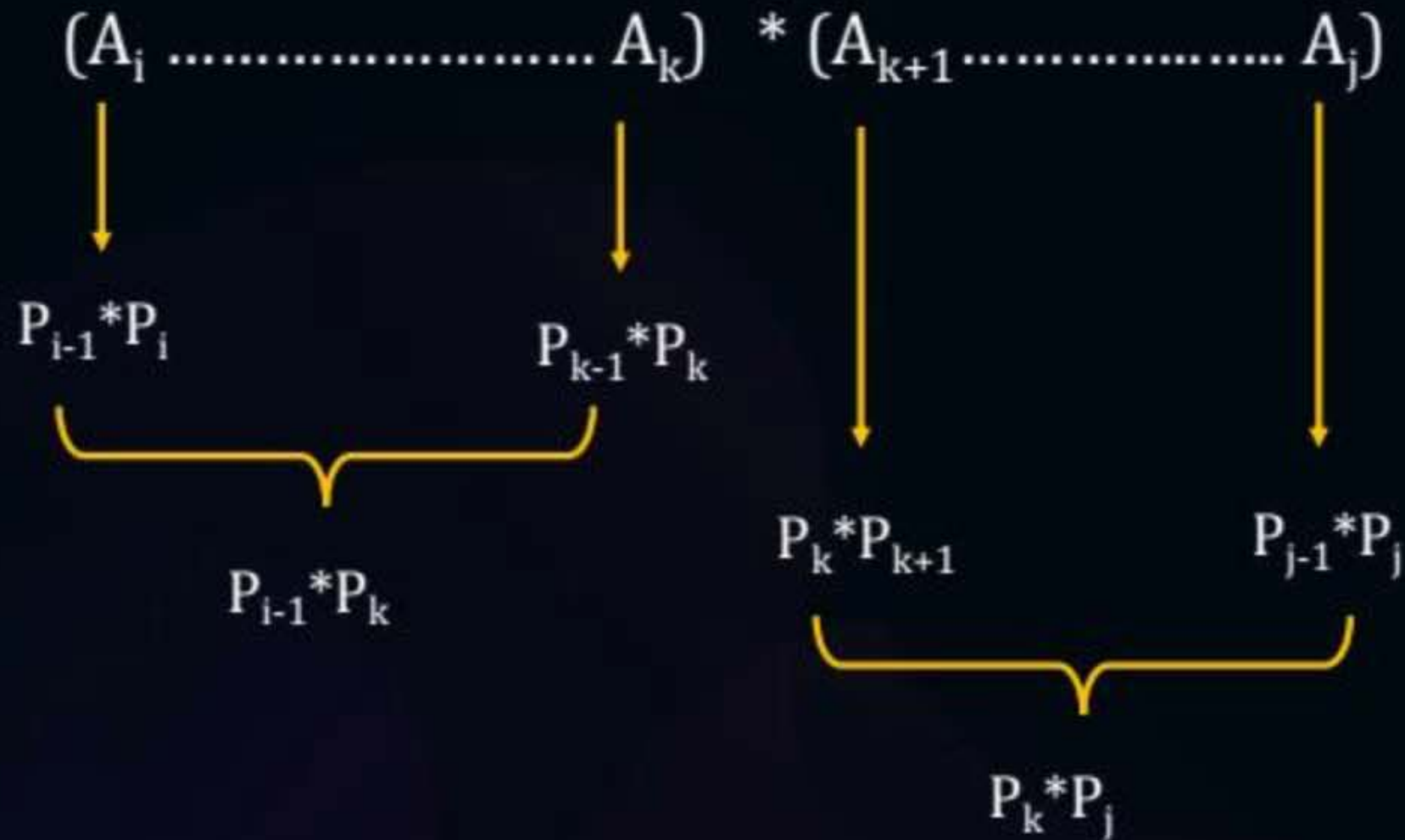


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HW

$$M[i, j] = \min \{m[i, k] + m[k+1, j] + P_{i-1} * P_k * P_j\}$$





2 mins Summary



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MCM



THANK - YOU